

Contains Duplicate (217)

Brute Force:- [1, 2, 3, 1]

we make freq arr
to find max ele

OR make the arr size constant

OR

sort arr & make two pointer which check adjacent value

[1, 4, 1, 1]

max

freq [max+1]

[0 1 2 ... 7]

After creating this

we placed frequency
in this arr

[0 1 2 3 4 ... 7]

After fill the freq we
check if any index
have more than one
So duplicate are
present

Time

- ① not able to
fill the freq
of negative value
because in constraints
there are value
which comes under
 $-10^9 \leq \text{val} \leq 10^9$

- ② And second is our
value is:-

So there are so
many wastage
of size

still there
are most of
storage are
wastage
in this sol

Problem:- but the tc is too
high in this
solution
S.C is $O(1)$.

Optimal Way:- [1, 2, 3000, 200, 1]

★

- ① Hash Set :- T.C is $O(1)$
S.C is $O(N)$

In this we are
creating hashset
and add the
value in the
set if the value
come again return true
other wise return
false.

- ② Optimal S.C is $O(1)$
T.C is $O(\log N)$

here we sort the
arr after that
all element are
adjacent to each
other with the
help of checking
the adjacent
we can easily
check there
are any
duplicate
present.

After sort

[1, 2, 200, 3000]

↓

compare the adjacent

If equal return true

or If no such adjacent is equal
return false